

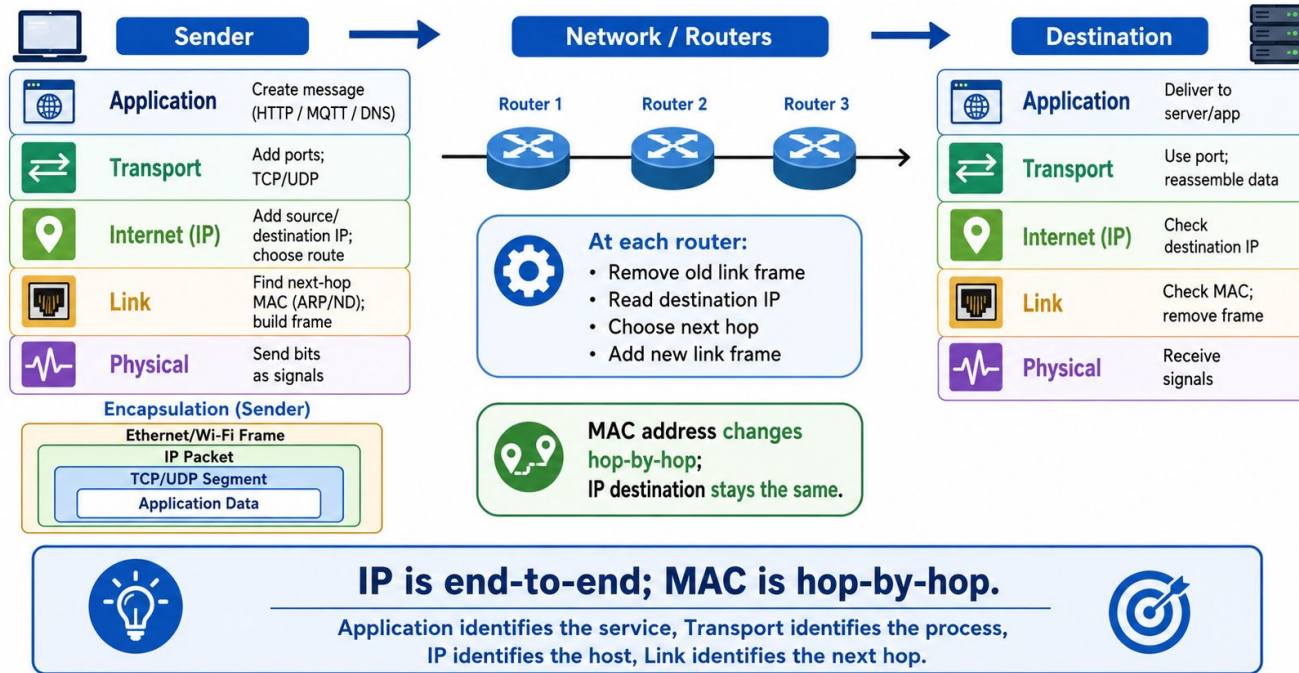
A decorative background featuring a network diagram. It consists of numerous nodes, represented by small circles, connected by thin lines. Some nodes are highlighted with a blue outline or a solid blue fill, while others are light gray. The network is distributed across the top-left and bottom-right corners of the image, framing the central text.

Communication Security

What?

- Identity (or Authentication): Ensuring that we are talking to the correct entity.
- Confidentiality: Ensuring that only our target is able to read the data being sent.
- Integrity: Ensuring that no one can modify the data being sent.

Lets Assume IP Stack!



Establishing Identity

- Host name => IP Address
- Need to contact DNS server:
 - `resolvectl status`
 - How do we ensure the DNS record is secure?
 - DNSSEC:
 - Records signed using CA.

Establishing Identity

- How do we know IP belongs to Host?
 - Public Key certificates.
- Before exchanging the information:
 - Get the certificate.
 - Verify the certificate.
 - If valid => continue

Confidentiality

- Encrypt all data.
 - Need to share keys.
 - Agree on cryptographic algorithm to use.
 - Start encrypting all the data before sending.
 - Decrypt all data before receiving.

Integrity

- Hash and Encrypt or Hash with a secret:
 - MAC (Message authentication code)

Communication Security

- Verify the identity.
- Agree on cryptographic algorithms.
- Establish shared secrets, i.e., encryption key and MAC key.
- Encrypt and hash all data with shared keys.
- Send/Receive application data.

Communication Security

Independent of the application protocol:

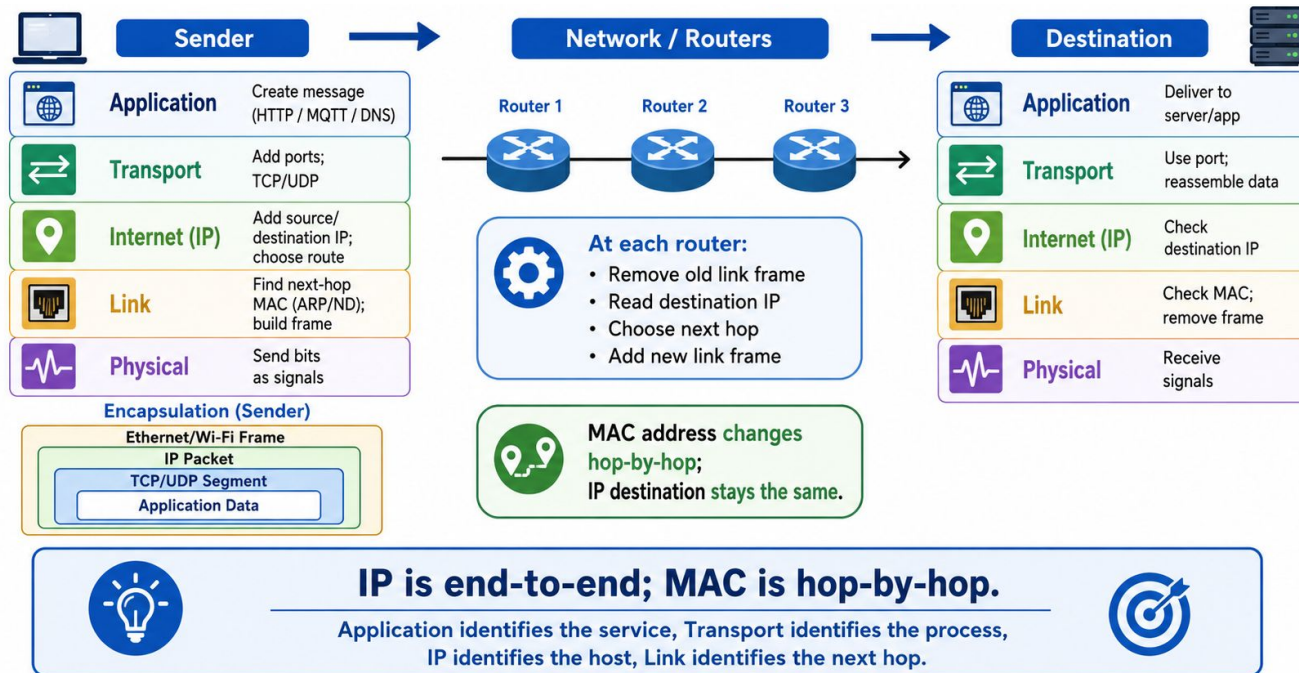
- Both HTTP and MQTT needs to follow the same mechanism.

Transport Layer Security

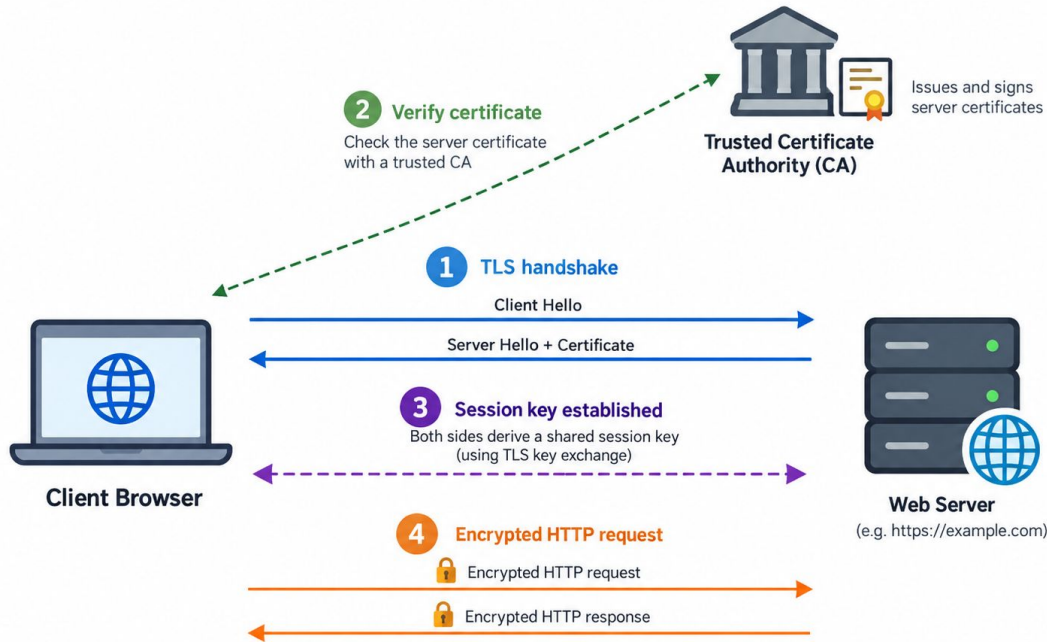
Adds **communication security** for any application data.

- Verify the identity.
- Agree on cryptographic algorithms.
- Establish shared secrets, i.e., encryption key and MAC key.
- Encrypt and hash all data with shared keys.
- Send/Receive application data.

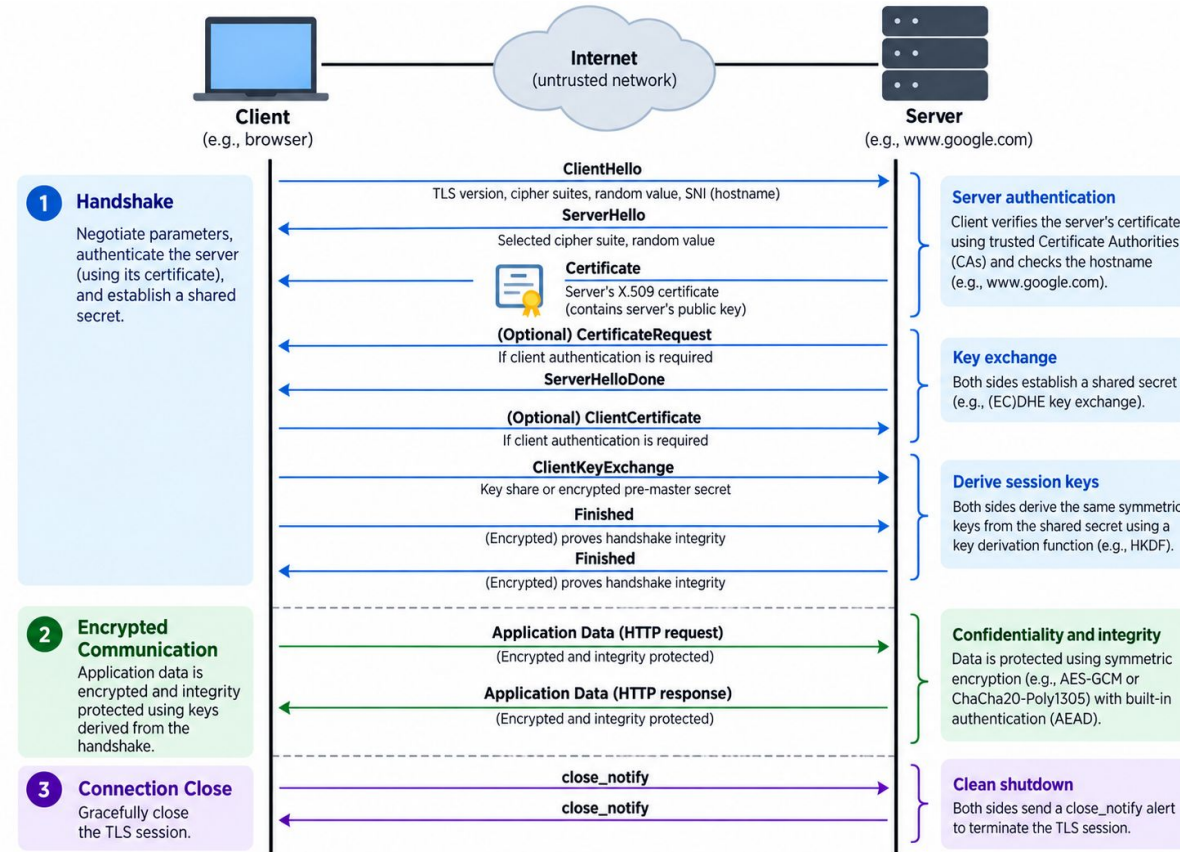
Where should we implement communication security?



HTTP + Communication Security = HTTPS



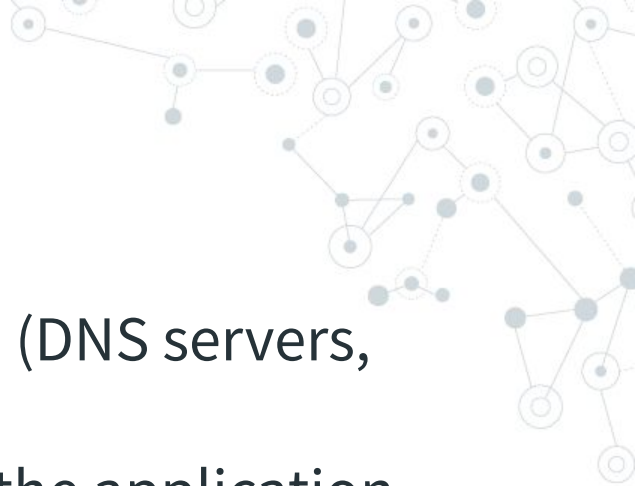
TLS



Application Protocol + Communication Security (S)

- HTTP (Port 80) -> HTTPS (Port 443)
- MQTT (Port 1883) -> MQTTS (Port 8883)
- etc.

Threats against TLS?



Best Practices

- Ensure your connection infrastructure (DNS servers, CA certificates) is secure.
- Ensure you always use (S) versions of the application protocol.
- Avoid using insecure protocols.